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NEP-2020 Readiness Index Construct Design – WIP (myelin internal brainstorming)

A. Weight distribution across Readiness Constructs (Total = 100)

This weighting reflects **"what unlocks change"**, not "what looks good early".

Readiness Construct	Weight	Why this weight (construct logic)
Intent Readiness	**20**	Necessary precondition. Without intent, nothing moves-but intent alone is cheap and over-reported.
Practice Readiness	**25**	Core enactment layer. This is where NEP intent becomes classroom reality through depth + reflection.
Capacity Readiness	**20**	Teacher cognition is the limiting reagent for diagnostic and adaptive practice.
System Readiness	**20**	Systems determine whether good practice sustains or collapses under load.
Ecosystem Readiness	**15**	Validation and early impact matter, but are downstream and sample-based in SRI v1.
Total	**100**	Balanced, non-dominant, trajectory-aware

Key principle respected:
No single construct exceeds 25 → avoids domination and false precision.

B. Mapping: 7 Index Signals → Construct → Weight → Value in Index

Map each signal explicitly, without early aggregation (Layer 1 logic preserved).

A. Intent Readiness (20)

Signal	Weight	Value it contributes
Leader Orientation (FP1–FP5)	**10**	Sets priority climate, legitimises NEP-aligned risk-taking.
Teacher Orientation (FP1–FP5)	**10**	Reveals where teachers naturally place attention before behaviour changes.

Why split evenly:
Misalignment (leader high, teacher low or vice versa) is itself a diagnostic signal, not noise.

B. Practice Readiness (25)

Signal	Weight	Value it contributes
Practice Depth (D1–D4 across FPs)	**15**	Primary evidence of enactment maturity (procedural → adaptive).
Reflection & Practice Trajectory	**10**	Distinguishes static practice from learning practice. Signals movement.

Why this is the heaviest construct:
NEP-2020 lives or dies here. Depth + reflection is where "learn to learn" conditions emerge.

C. Capacity Readiness (20)

Signal	Weight	Value it contributes

Teacher Cognitive Capacity (Pre/Post)	**20**	Determines ceiling for diagnostic reasoning, abstraction, and adaptation.

Why single-signal but high weight:
Cognition is not inferred- it is directly measured and strongly predictive of depth progression.

D. System Readiness (20)

Signal	Weight	Value it contributes
Teacher Enablement (Experience lens)	**10**	Whether systems reduce friction or add cognitive load.
Leadership Routines (Structure lens)	**10**	Whether practice is protected, reviewed, and normalised institutionally.

Important nuance:
Strong systems with weak practice ≠ readiness
Strong practice with weak systems = fragile readiness

SRI surfaces this tension deliberately.

E. Ecosystem Readiness (15)

Signal	Weight	Value it contributes
Parent Validation Signals	**7**	Confirms whether learning communication and trust are felt externally.
Early Student Signals (sample-based)	**8**	Indicates exposure to cognitive demand, not achievement.

Why slightly more weight to student than parent:
Students reflect classroom conditions more directly; parents validate alignment and credibility.

1. Intent Readiness

Purpose

Intent Readiness captures whether a school's leadership and teachers demonstrate a shared, structurally complete orientation toward NEP-2020, sufficient to support meaningful implementation in practice.

Intent Readiness **does not** assess behaviour or outcomes.
It surfaces where attention naturally goes when leaders and teachers think about teaching, learning, and school priorities.

Signals Contributing to Intent Readiness

Intent Readiness is derived from **two independent raw signals**:

1. **Leader FP Orientation**
2. **Teacher FP Orientation**

Each signal is analysed separately and then synthesised to produce the construct value.

Maximum construct weight: 20

- Leader Intent: 10
- Teacher Intent: 10

Raw signal definition

FP Orientation Responses

Respondents (leaders or teachers) answer a structured set of questions, where each response option is mapped to one of the five NEP-2020 Foundational Principles (FPs):

- FP1: Every Child is Unique

- FP2: Holistic, Experiential Learning
- FP3: Teacher as Reflective Practitioner
- FP4: Assessment for Learning
- FP5: Collaboration & Community

Each selection is treated as an **attention signal**, not a correctness judgment.

Step 1: FP-wise Attention Density Vector (ADV)

Definition

For a given group (leaders or teachers), responses are converted into an **FP-wise Attention Density Vector**:

ADV=[d1,d2,d3,d4,d5]

where:

di=Number of selections mapped to FPi / Total selections

Properties:

- $\sum di=1$
- Each $di \in [0,1]$

Why this is used

- Removes sample size effects
- Preserves **relative emphasis** across principles
- Avoids over-rewarding volume of responses

The ADV is the **base representation** of intent.

Example: Teacher FP Attention Density (ADV)

Assume 10 questions × 20 teachers = 200 selections.

FP	Count	Density
FP1	60	0.30
FP2	40	0.20
FP3	70	0.35
FP4	20	0.10
FP5	10	0.05
Total	200	**1.00**

So teacher ADV =
[0.30, 0.20, 0.35, 0.10, 0.05]

This vector is **the signal**.

Step 2: Breadth Component (Coverage)

Definition

Breadth measures whether intent meaningfully spans NEP principles, or is narrowly concentrated.

A minimum attention threshold τ is defined (default: 0.10).

τ may be adjusted during sensitivity testing, but is fixed within a given SRI cycle.

Coverage=Number of FPs where $d_i \geq \tau / 5$

Range: [0,1]

Interpretation

- High coverage → multiple principles receive meaningful attention
- Low coverage → blind spots in NEP intent

Coverage protects against **partial NEP adoption**.

Step 3: Coherence Component (Balance)

Definition

Coherence measures whether NEP intent is **structurally complete enough to survive real classroom complexity**, rather than collapsing into one dominant belief.

Balance does not imply equal emphasis across principles; it indicates that no principle is systematically neglected.

Coherence is operationalised as the **absence of extreme skew** in the Attention Density Vector (**ADV**)

Computation

The maximum variance (0.16) is a theoretical bound determined by the number of Foundational Principles (5), corresponding to complete concentration of intent in a single principle. It is not derived from empirical data.

1. Reference distribution (structural completeness baseline):

U=[0.2,0.2,0.2,0.2,0.2]

2. Compute population variance of ADV:

$$Var = \frac{1}{5}\sum_{i=1}^{5}(di - 0.2)^2$$

Example

FP	di	di − 0.20	(di − 0.20)²
FP1	0.30	+0.10	0.0100
FP2	0.20	0.00	0.0000
FP3	0.35	+0.15	0.0225
FP4	0.10	−0.10	0.0100
FP5	0.05	−0.15	0.0225

Sum			**0.0650**

Variance = 0.0650 / 5 = 0.013

3. Compute maximum possible variance (all attention on one FP):

   Maximum variance occurs when **all attention collapses into one FP**:

[1.0, 0, 0, 0, 0]

$$Var_{max} = 0.16$$

Example

Compute variance:

FP	di	di − 0.20	$(di - 0.20)^2$
FP1	1.00	+0.80	0.6400
FP2	0.00	−0.20	0.0400
FP3	0.00	−0.20	0.0400
FP4	0.00	−0.20	0.0400
FP5	0.00	−0.20	0.0400
Sum			**0.8000**

Max Variance = 0.8000 / 5 = 0.16 (this is worst case tunnel vision)

4. Normalise variance:

$$Var_{norm}=Var/Var_{max}$$

   Example

   Normalised Variance = 0.013 / 0.16 ≈ 0.081

5. Convert to Balance score:

$$\text{Balance} = 1 - \text{Var}_{\text{norm}}$$

Range: [0,1]

Interpretation

- Balance ≈ 1 → focused yet structurally complete intent
- Balance → 0 → tunnel vision

Coherence protects against **ideological monoculture**, even when alignment exists.

Step 4: Alignment Component (Leader–Teacher Alignment)

Definition

Alignment measures whether leaders and teachers are oriented toward **the same NEP principles**, even if their emphasis differs. It measures directional similarity of attention, not agreement on intensity, quality, or practice.

Given:

- ADV_L = Leader Attention Density Vector
- ADV_T = Teacher Attention Density Vector

Alignment is computed using cosine similarity:

$$\text{Alignment} = ADV_L \cdot ADVT / \|ADV_L\| \, \|ADV_T\|$$

Range: [0,1]

Cosine similarity answers: "If we ignore scale, how similarly are attention weights distributed?"

Alignment is symmetric; leader–teacher and teacher–leader comparisons yield the same value.

Example

Leader ADV (L): = [0.25, 0.25, 0.20, 0.20, 0.10]

Teacher ADV (T): = [0.30, 0.20, 0.35, 0.10, 0.05]

Dot product (shared emphasis)

L·T=(0.25×0.30)+(0.25×0.20)+(0.20×0.35)+(0.20×0.10)+(0.10×0.05)

0.075 + 0.050 + 0.070 + 0.020 + 0.005 = 0.220

Vector magnitudes:

$$\|L\| = \sqrt{0.25^2 + 0.25^2 + 0.20^2 + 0.20^2 + 0.10^2} = \sqrt{0.215} = 0.464$$

$$\|T\| = \sqrt{0.30^2 + 0.20^2 + 0.35^2 + 0.10^2 + 0.05^2} = \sqrt{0.265} = 0.515$$

Cosine similarity (Alignment)

Alignment= L·T/ $\|L\| \times \|T\| = 0.220/0.464 \text{ X } 0.515 = 0.92$

Interpretation

- Alignment ≈ 1 → shared intent direction
- Low alignment → fragmentation between leadership and classroom priorities

Alignment protects against **disconnect between vision and practice**.

Step 5: Intent Signal Contribution (per group)

For leaders and teachers separately, an **Intent Signal Score** is computed as:

IntentScore = Coverage × Balance × Alignment

This multiplicative structure ensures that:

- No single strength can compensate for a structural weakness
- Narrow, skewed, or misaligned intent is penalised

Step 6: Construct Aggregation

- Leader Intent Score is mapped to **0–10**
- Teacher Intent Score is mapped to **0–10**

Intent Readiness=Leader Intent + Teacher Intent

Maximum = 20

An increase in Intent Readiness represents

If Intent Readiness increases, it must correspond to observable changes such as:

- Attention spreading to previously neglected NEP principles
- Reduction in single-principle dominance
- Greater convergence between leadership and teacher priorities

If such changes cannot be articulated, the computation is considered invalid.

Design Guardrails (explicit)

- Intent Readiness does not imply implementation quality
- High intent with low practice is a valid and expected state
- Alignment without coherence is treated as constrained readiness

2. Practice Readiness

Step 1: Practice Depth Distribution Matrix (PDDM)

Purpose of Step 1

The Practice Depth Distribution Matrix captures how NEP intent is enacted in classrooms, by jointly representing:

- Which Foundational Principles (FP1–FP5) are being enacted
- At what depth (D1–D4) those practices operate

This step does not judge quality or effectiveness.
It establishes a structural map of classroom practice.

Raw Signal Definition (Input)

Practice Depth Responses

Teachers respond to classroom-practice items where each response is mapped to:

- One Foundational Principle (FP1–FP5)
- One Practice Depth band (D1–D4)

Practice Depth Bands

Depth	Meaning (operational)
D1	Procedural / rule-following
D2	Responsive / situational adjustment
D3	Diagnostic / evidence-based decision
D4	Adaptive / intentional redesign

Each response represents **how practice is enacted**, not whether it is correct or desirable.

Step 1.1: Constructing the Matrix

For a given school (or teacher cohort), responses are aggregated into a **5 × 4 matrix**:

PDDM=

$$\begin{matrix} FP1D1 & FP1D2 & FP1D2 \vdots\vdots \\ FP2D1 & FP2D2 & FP2D2 \vdots\vdots \\ FP3D1 & FP3D2 & FP3D3 \vdots\vdots \end{matrix}$$

Where each cell contains the **count of responses** mapped to that FP–Depth combination.

Step 1.2: Normalising the Matrix

To remove sample-size effects, the matrix is converted into **proportions**.

Let total practice responses = N

$$\mathrm{Pij} = \mathrm{FPi\ at\ depth\ Dj}/N$$

Properties:

- All cells Pij ∈ [0,1]
- Sum of all cells = 1

This produces the **Practice Depth Distribution Matrix (normalised)**.

Worked Example (Illustrative)

Assume:

- 5 FPs
- 4 Depth bands
- Total responses = 200

Raw counts (example)

FP \ Depth	D1	D2	D3	D4	Row Total
FP1	12	10	6	2	30
FP2	14	12	6	3	35
FP3	20	18	12	5	55
FP4	18	16	8	3	45
FP5	15	10	6	4	35
Total					**200**

Normalised PDDM (selected cells)

FP \ Depth	D1	D2	D3	D4
FP1	0.06	0.05	0.03	0.01
FP3	0.10	0.09	0.06	0.03
FP5	0.075	0.05	0.03	0.02

(Full matrix sums to 1)

Why this matrix is necessary and cannot be skipped

What PDDM preserves

- Depth distribution **(D1 vs D3 is visible)**
- Principle-wise enactment **(which FPs stay procedural)**
- Structural gaps **(e.g., FP4 never reaches D3/D4)**

What it deliberately avoids

- No averaging across depths
- No collapsing FP differences
- No premature "score"

This is the **practice analogue** of the Attention Density Vector in Intent Readiness.

Conceptual Interpretation

From the matrix alone, we can already observe:

- Practices concentrated in **D1/D2** → enactment without diagnosis
- Sparse **D3/D4** cells → limited learner-responsive practice
- Uneven FP rows → partial NEP enactment

But **no judgement is made yet**.

Design Guardrails

- The matrix is **descriptive, not evaluative**
- High D1 presence is expected in early stages
- D4 scarcity is normal and not penalised here
- Interpretation requires later steps (Depth Effectiveness + Trajectory)

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